4 (a) (i) F / A B1 [1]

(ii) $\Delta L / L$ B1 [1]

(iii) allow $FL / A\Delta L$ B1 [1]

(iv) allow $\rho L / A$ or $\rho(L + \Delta L) / A$ B1 [1]

(b) (i) $\Delta L = FL / EA$
$= (30 \times 2.6) / (7.0 \times 10^{10} \times 3.8 \times 10^{-7})$ M1
$= 2.93 \times 10^{-3}$ m $= 2.93$ mm A0 [1]

(ii) $\Delta R = \rho\Delta L / A$ C1
$= (2.6 \times 10^{-8} \times 2.93 \times 10^{-3}) / (3.8 \times 10^{-7})$
$= 2.0 \times 10^{-4}\ \Omega$ A1 [2]

(c) change in resistance is (very) small M1
so method is not appropriate A1 [2]

5 (a) when a wave passes through a slit / by an edge M1